

MediPredixt

37 questions

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Q1

architecture | senior

In `app.py`, the `local_css()` function injects CSS via `st.markdown()` with embedded style tags. Why did you choose this approach instead of using Streamlit's built-in theming system or external CSS files? What are the trade-offs of this decision in terms of maintainability and performance?

Q2

architecture | senior

The application uses Streamlit's multi-page feature with separate files in the `pages/` directory. How does this architecture handle shared state between pages, and what limitations does this impose on the application's scalability?

Q3

implementation | senior

In the prediction pages (`pages/cancer.py` and `pages/diabetes.py`), the code loads model files using `pickle.load()` directly in the prediction function. What are the performance implications of this approach, and how would you optimize it for production use?

Q4

implementation | mid-level

The PDF report generation in the prediction pages uses the FPDF library. Walk through how you would implement input validation for the user-provided data that gets included in these reports. What specific validations would you add for the cancer prediction inputs?

Q5

tradeoffs | staff

The application commits serialized model files (.pkl) directly to the repository. Compare this approach to alternative model deployment strategies in terms of security, maintainability, and scalability. When would you choose each approach?

Q6

tradeoffs | senior

The application uses Streamlit's monolithic execution model where the entire script re-runs on each interaction. For a healthcare application making predictions, what are the trade-offs of this approach versus a traditional request-response web framework like Flask or FastAPI?

Q7

scalability | staff

The application loads machine learning models directly from .pkl files on disk. What are the scalability limitations of this approach, and how would you redesign the system to handle 10,000 concurrent users making predictions?

Q8

scalability | senior

The PDF report generation happens client-side in the Streamlit application. What are the scalability challenges of this approach, and how would you redesign it to handle increased load?

Q9

designPatterns | senior

The application shows signs of the Template Method pattern in the prediction workflows. How would you explicitly implement this pattern to standardize the prediction flow across different disease modules while allowing for disease-specific customizations?

Q10

security | senior

The application passes user inputs directly to the machine learning models without validation. What security vulnerabilities does this introduce, and how would you implement comprehensive input validation for the cancer prediction module?

Q11

architecture | staff

The application uses a monolithic architecture with all components bundled together. How would you decompose this into microservices while maintaining the current user experience? What specific services would you create and how would they communicate?

Q12

implementation | mid-level

In `app.py`, the navigation links in the `navbar` are hardcoded. How would you refactor this to dynamically generate navigation items based on available pages in the `pages/` directory? What benefits would this provide?

Q13

tradeoffs | senior

The application generates PDF reports client-side using FPDF. Compare this approach to server-side PDF generation in terms of security, performance, and user experience. When would you choose each approach?

Q14

scalability | senior

The application uses Streamlit's session state for shared data between pages. What are the scalability limitations of this approach, and how would you redesign the state management to support a distributed deployment?

Q15

designPatterns | mid-level

The application violates the DRY principle with duplicated CSS styling across multiple files. How would you refactor the styling to eliminate duplication while maintaining the current visual design?

Q16

security | senior

The application commits sensitive model files (.pkl) directly to the repository. What security risks does this introduce, and how would you implement a secure model deployment pipeline?

Q17

architecture | senior

The application has a large `app.py` file handling multiple responsibilities. How would you decompose this into smaller, focused modules while maintaining the Streamlit application structure?

Q18

implementation | mid-level

The prediction pages show loading indicators using Streamlit's `st.spinner()`. How would you implement more sophisticated loading states that provide better user feedback, especially for long-running predictions?

Q19

tradeoffs | senior

The application uses Streamlit's built-in UI components rather than custom HTML/CSS. What are the trade-offs of this approach in terms of customization, performance, and developer experience?

Q20

scalability | staff

The application currently loads all models into memory at startup. How would you redesign this to support a growing number of disease prediction models without increasing memory usage proportionally?

Q21

architecture | mid-level

The application's data flow isn't explicitly documented. Map out the complete data flow for a cancer prediction request, identifying all components and transformations the data passes through from user input to PDF report generation.

Q22

implementation | senior

The application uses `pickle` to serialize machine learning models. What are the security implications of using pickle, and what alternative serialization formats would you consider for production deployment?

Q23

tradeoffs | senior

The application generates PDF reports immediately after predictions. What are the trade-offs of this synchronous approach versus an asynchronous report generation system? How would you implement the latter?

Q24

scalability | senior

The application's Streamlit architecture re-runs the entire script on each interaction. What caching strategies would you implement to optimize performance for frequent user interactions?

Q25

architecture | senior

The application doesn't have explicit error handling for model prediction failures. Design a comprehensive error handling strategy that covers model loading, prediction execution, and result interpretation failures.

Q26

implementation | mid-level

The application uses hardcoded model paths. How would you refactor this to make the application more flexible and easier to deploy in different environments?

Q27

tradeoffs | mid-level

The application uses FPDF for PDF generation. Compare this to alternative PDF generation libraries in terms of features, performance, and ease of use for this specific use case.

Q28

scalability | staff

The application's memory usage scales linearly with the number of concurrent users. What strategies would you implement to reduce the memory footprint per user?

Q29

architecture | senior

The application has a flat file structure with no clear module boundaries. How would you reorganize the codebase to better reflect the application's architecture and improve maintainability?

Q30

implementation | senior

The application uses Streamlit's `st.markdown()` for displaying information. How would you implement a more structured approach to content rendering that supports localization and accessibility requirements?

Q31

tradeoffs | senior

The application uses client-side PDF generation. What are the trade-offs of generating PDFs on the client versus using a dedicated PDF generation service, in terms of cost, reliability, and user experience?

Q32

scalability | staff

The application's Streamlit architecture makes it difficult to implement rate limiting. How would you design rate limiting for the prediction endpoints while maintaining the current user experience?

Q33

architecture | senior

The application doesn't have a clear separation between UI and business logic. How would you refactor the cancer prediction page to implement a proper Model-View-Controller pattern?

Q34

implementation | senior

The application uses `st.session_state` for sharing data between pages. How would you implement a more robust state management system that supports complex data types and better error handling?

Q35

tradeoffs | senior

The application uses pre-trained models loaded from disk. What are the trade-offs of this approach versus training models on-demand or using a model serving API?

Q36

scalability | staff

The application's Streamlit architecture makes it difficult to implement background jobs. How would you design a system to handle long-running tasks like model training or report generation without blocking the user interface?

Q37

architecture | senior

The application doesn't have a clear testing strategy. Design a comprehensive testing strategy that covers unit tests, integration tests, and end-to-end tests for the cancer prediction feature.